



Ulimpiad n Tefriqt n Ugafa n Tselkimt Tamenzut 2025

Target Rotation

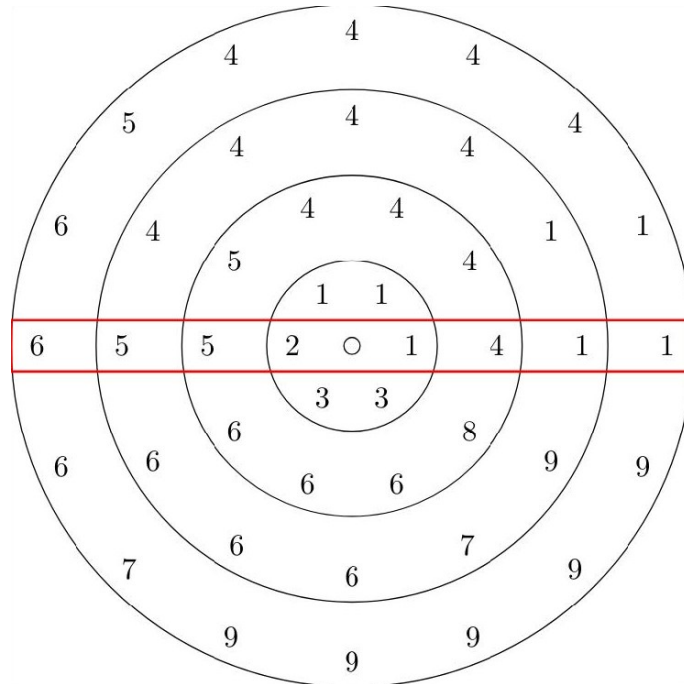
Tilist n wakud: 1.5 secondes

Tilist n tkatut: 256 MB

Deg temdint n **Djelfa**, **Ahmad Karawita** akked **Hamoud Habibi Hamoud** hemlen ad uraren yiwen n urar s **idebsiyen imdewranen**. Llan n n idebsiyen, ismawen-nsen $d_1, d_2, \dots, d_n$, anda d_i d adebsi yesεan lqis i . Yal adebsi d_i yesεa uṭṭunen $0, 1, \dots, 9$ ttwareṭṭben yerna nura-ten **mgal tama n ssaεa**, seg umekkan yellan deg lḡiha tayeffust akk n uṭṭensi. Yal uṭṭun j γef uṭṭensi d_i yettwawεawed $a_{j,i}$ n yiberdan, uṭṭun n wuṭṭunen deg yal adebsi d amsas (pair).

Idεbsiyen rsen wa γef wa s yiwet n tlemmast, ideg d_1 yella deg usawen yerna d_n deg ukessar. Imyuraren zemren ad dewwren yal adebsi s yiwen n uṭṭun akken ma bγan, s tlelli deg lḡiha i bγan.

Iswi-nsen ad d-suffγen yiwen n lkud deg ujerriḍ alemmas, am deg tugnat. Ajerriḍ alemmas deg-s $2n$ n uṭṭunen: yal adebsi deg-s sin, yiwen deg lḡiha tayeffust, wayeḍ deg tazelmaḍt.



Ccγel

Akken ad ilin $x_1, x_2, \dots, x_q$ q n lkudat, yal yiwen deg-s $2n$ n uṭṭunen, Af-d **aḥal n yiberdan ilaq ad ndewwer** akken ad d-nsufeγ yal lkud. Ma ur nezmir ara, sufficeγ-d -1 . Yal asteqsi d ilelli γef wiyad, deg wawalen-nniḍen, deg yal asteqsi, mi ara nfakk, idebsiyyen ttuγalen akken i llan qbel.

Anekcum

1. Utɥun n ($1 \leq n \leq 10^5$): utɥun n iɗebsiyen.
2. Tafelwit n $10 \times n$ n utɥunen, anda $a_{j,i} \leq 10^9$ ($0 \leq j \leq 9, 1 \leq i \leq n$) yeqqar-d aɗal n yiberdan i yettɛiwid utɥun j γef uɗebsi d_i . Utɥun n wutɥunen deg yal aɗebsi d amsas (pair).
3. Utɥun q ($1 \leq q \leq 2 \times 10^6$): utɥun n lkudat.
4. q n lkudat $x_1, x_2, \dots, x_q$, yal yiwen lqis-is $2n$.
5. $nq \leq 7 \times 10^6$

Tuffɣa

I yal lkud x_k ($1 \leq k \leq q$), sufferɣ-d:

- Deg ujerrid amenzu, sufferɣ-d POSSIBLE ma nezmer ad d-nsuffer lkud x_k .
- Deg ujerrid wis sin, **aɗal n yiberdan ilaq ad ndewwer** akken ad d-yeffer lkud x_k , ma ur nezmir ara, sufferɣ-d -1.

Ma yella ajerriɗ amenzu yesseha maca wis sin ala, ad d-tawid nnus n tneqqiɗin.

Tiɣuriwin

- $1 \leq n \leq 10^5$
- $0 \leq a_{j,i} \leq 10^9$
- $1 \leq q \leq 2 \times 10^6$
- $nq \leq 7 \times 10^6$

Iluɣma imezɣanen

Alaɣmu	Tineqqiɗin	Tiɣuriwin
1	6	$n = q = 1$
2	6	Yal aɗebsi deg-s yiwen n utɥun kan
3	10	Pour tout $i, \sum_{j=0}^9 a_{j,i} \leq 2$
4	8	$n = 1$
5	20	I yal $i, j, x, y : a_{i,j} = a_{x,y}$
6	30	$nq \leq 10^5$
7	20	-

Imedyaten

Anekcum

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4
0 3 1 2 0 0 0 0 0 0
0 0 0 0 4 2 3 0 1 0
0 2 0 0 4 1 3 1 0 1
0 2 0 0 4 1 3 1 0 5
4
65521411
65521419
75521414
75521410

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Weyi d amedya i yellan deg usebter 1.

Tuff γ a

POSSIBLE
0
POSSIBLE
1
POSSIBLE
2
IMPOSSIBLE
-1

Anekcum

1
4 3 2 3 1 1 4 3 9 2
1
71

Tuff γ a

POSSIBLE
4